

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 09



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TOPICS TO BE COVERED

01 Basics of Generating function

02 Problems on Generating function

COMBINATORICS



Generating Function:

$a_0, a_1, a_2, a_3, a_4, \dots, \infty$

$$G(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots$$

$$G(x) = \sum_{i=0}^{\infty} a_i \cdot x^i$$

COMBINATORICS



$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 1 & \dots & \infty \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & & \\ a_0 & a_1 & a_2 & a_3 & a_4 & & \end{array}$$

$$G(x) = \frac{1}{1-x}$$

$$G(x) = a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \dots$$

$$= 1 \cdot x^0 + 1 \cdot x + 1 \cdot x^2 + 1 \cdot x^3 + \dots$$

$$G(x) = 1 + x + x^2 + x^3 + x^4 + \dots$$

COMBINATORICS



$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots$$

$$a = 3$$

$$\frac{1}{1-3x} = 1 + (3x) + (3x)^2 + (3x)^3 + \dots$$

$$\frac{1}{1-ax} = 1 + ax + (ax)^2 + (ax)^3 + \dots$$

$$a = 2$$

$$\frac{1}{1-2x} = 1 + (2x) + (2x)^2 + (2x)^3 + \dots$$

$$\frac{1}{1+ax} = 1 - ax + (ax)^2 - (ax)^3 + (ax)^4 + \dots$$

COMBINATORICS



$a_2 \quad a_3$
 $0, 0, 0, 6, -6, 6, -6, \dots$
 $a_0 \uparrow \quad a_1 \downarrow$

$$G(x) = \frac{6x^3}{1+x}$$

$$G(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$$

$$= 6x^3 - 6x^4 + 6x^5 - 6x^6 + \dots$$

$$= 6x^3(1 - x + x^2 - x^3 + x^4 - \dots) = \frac{6x^3}{1+x}$$

COMBINATORICS



$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$\frac{1}{(1-x)^2} = 0 + 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \dots = \sum_{i=0}^{\infty} (i+1) \cdot x^i$$

$$d\left(\frac{1}{n}\right) = -\frac{1}{n^2}.$$

$$d\left(\frac{1}{1-n}\right) = \frac{-1}{(1-n)^2} \frac{d(-n)}{dn}$$

$$= \frac{-1}{(1-n)^2} \cdot (-1) = \frac{1}{(1-n)^2}$$

COMBINATORICS



$$G(x) = \frac{1}{1-x^2}$$

$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 1 & 0 & \dots \\ a_0 & a_1 & a_2 & a_3 & a_4 & a_5 & \end{array}$$

$$G(x) = a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

$$= 1 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 + 0 \cdot x^3 + 1 \cdot x^4 + 0 \cdot x^5 + 1 \cdot x^6 + \dots$$

$$= 1 + x^2 + x^4 + x^6 + x^8 + \dots$$

$$= 1 + x^2 + (x^2)^2 + (x^2)^3 + (x^2)^4 + \dots$$

COMBINATORICS



$$\begin{matrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ 0, & 0, & 1, & a, & a^2, & a^3 \dots \end{matrix}$$

$$G(x) = \frac{x^2}{1-ax}$$

$$a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + a_4x^4$$

$$G(x) = \frac{x^2}{1-ax}$$

$$G(x) = \underline{0} \cdot x^0 + \underline{0} \cdot x^1 + 1 \cdot x^2 + a \cdot x^3 + a^2 x^4 + a^3 x^5 \dots$$

$$= x^2 + ax^3 + a^2 x^4 + \dots$$

$$x^2 (1 + ax + (ax)^2 + (ax)^3 + (ax)^4 \dots)$$

COMBINATORICS



$$G(x) = \frac{x^2}{1-ax}$$

$$= x^2 [1 + ax + (ax)^2 + (ax)^3 + \dots]$$

$$= \underline{1} \cdot \underline{x^2} + \underline{a} \underline{x^3} + \underline{a^2} \underline{x^4} + \dots$$

$$G(x) = \underline{a_0} x^0 + a_1 x^1 + \underline{a_2} x^2 + \underline{a_3} x^3 + a_4 \underline{x^4} + \dots$$

$$0, 0, 1, a, a^2, a^3, \dots$$

COMBINATORICS



$$G(x) = \frac{x^3}{1-x^2}$$

$$\frac{1}{1-x^2} = 1 + x^2 + (x^2)^2 + (x^2)^3 + (x^2)^4 + \dots$$

$$= x^3 [1 + x^2 + x^4 + x^6 + x^8 \dots]$$

$$G(x) = \underline{x^3} + \underline{x^5} + x^7 + x^9 + \dots$$

$$a_0 = 0$$

$$a_4 = 0$$

$$a_1 = 0$$

$$a_5 = 1$$

$$a_2 = 0$$

$$a_6 = 0$$

$$a_3 = 1$$

$$a_7 = 1$$

$$a_0 x^0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \underline{a_4 x^4} + \underline{a_5 x^5} + \dots$$

0, 0, 0, 1, 0, 1, 0, 1

COMBINATORICS



$$1) G(x) = \frac{x^4}{1-x} \quad 2) G(x) = \frac{1}{1+3x}$$

$$3) G(x) = \frac{1}{1-x} + (3x^7 - 11)$$

$$\frac{1}{1-x} + (3x^7 - 11)$$

$$(1 + x + x^2 + x^3 + x^4 + \dots) + (3x^7 - 11)$$

$$(1 - 11) + x + x^2 + x^3 + \dots + (x^7 + 3x^7) + x^8 + x^9 + x^{10} + \dots$$

$$-10 + x + x^2 + x^3 + \dots + 4x^7 + x^8 + \dots$$

$$a_0 = -10$$

$$a_7 = 4$$

$$a_i = 1 \quad (i \neq 0, 7)$$

COMBINATORICS

$$G(x) = \frac{1}{1-x} + (3x^7 - 11)$$

$$a_0 = -10$$

$$a_7 = 4$$

$$a_i = 1 \quad [i \neq 0, 7]$$

$$= [1 + \underline{x} + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + \dots] + [3x^7 - 11]$$

$$= (\underline{1 - 11}) + x + x^2 + x^3 + x^4 + x^5 + x^6 + (\underline{x^7 + 3x^7}) + x^8 + x^9 \dots$$

$$= -10 + x + x^2 + \dots + x^6 + 4x^7 + x^8 + x^9$$

$$a_0 \quad a_1 \quad a_2$$

$$a_7$$

$$a_8$$

$$a_9$$

COMBINATORICS



$$G(x) = (1-x)^{-2} = \sum a(i) x^i$$

$$g(i) \quad \text{A) } 1 \quad \text{C) } 2i$$

$$\text{b) } i+1 \quad \text{d) } 2i+1.$$

